

Solution Outputs — Team AuraQu

Challenge Statement §5, outputs 2 and 3. The connectivity matrices (output 1) are supplied separately as `Connectivity_Matrices.csv`; the methodological argument is Section 2 of the Concept Proposal.

1. Hit List — top five predicted allosteric residues per target

residue_number is the PDB number as deposited and is the one to quote. **residue_index** is the 0-based position in the connectivity matrix. They differ for three of the four targets, because PDB numbering does not start at zero and is not always contiguous.

The same five, as matrix indices:

KRAS_G12C is the trap: its structure genuinely numbers from 0, so its two rows coincide. Code tested only on KRAS will silently use the wrong column on the other three.

Read the verdicts with the list

For the three targets where a drug-bound structure exists to check against, our own validation returns **NO_SIGNAL_IN_APO**: the score's confidence interval overlaps that of the best trivial baseline computed on the same structure.

Every interval includes zero. **We submit the five as required and state plainly that we cannot certify them** — the conclusion our proposal reaches about the field's published numbers, applied to our own.

1NKP (c-Myc) has no drug-bound structure in the PDB, so no ground truth exists to score against. Its five are a genuinely prospective prediction, labelled **unverified**, from a four-operator consensus.

2. Known limitations, measured

Where the five actually sit, per target. Measured here rather than asserted: heavy-atom minimum distance, altloc A only, hydrogens excluded; the truth pocket is every residue within 5.0 Å of the validation ligand (MYR in 1OPL, sotorasib MOV in 6OIM, mavacamten XB2 in 8QYR). Reproducible with `.ai/tools/verify_hit_list_distances.py`.

Three readings the Concept Proposal compresses into one line. **BCR-ABL1 and cardiac myosin miss completely** — nothing we submit comes within 16 Å of

the validated pocket, while everything we submit sits 4–12 Å from the orthosteric ligand. **KRAS is not a partial success but an unresolvable case:** its switch-II pocket abuts the nucleotide, so residue 33 lies 1.3 Å from the pocket and 4.3 Å from GDP·Mg — at this separation the two labels cannot be told apart, which is the confound itself rather than a recovery. **BCR-ABL1's #5 is residue 338**, the gatekeeper threonine (T315 in Abl-1a numbering, 338 - 19), 4.2 Å from the ATP-site inhibitor and 16.7 Å from the myristoyl pocket the challenge names as the target.

Against our own site-level detection threshold of Jac-card ≥ 0.3 , that is **0 of 3**.

Each hit list is one draw. Ten genuine G12C apo structures span AUC **0.408–0.595** (median 0.482); substituting the cardiac apo structure moved AUC by **0.27**. A different, equally defensible apo deposition would give a different five.

A classical cavity detector beats our walk on one of three targets and loses on another. Re-run fresh against the currently shipped structures: fpocket 0.860 vs our 0.541 on BCR-ABL1; 0.420 vs 0.514 on KRAS G12C; 0.535 vs 0.548 on cardiac myosin. We report this because fpocket is a stage inside our own pipeline, and because it is evidence the benchmark is easier than it looks.

Pooled AUC across proteins carries a protein-identity floor. A score with all within-structure position information destroyed — every residue replaced by its own protein's mean — reaches pooled **AUC 0.65**. Every AUC in this submission is computed per structure and averaged, never pooled; we audited that rather than assuming it.

Our own orthosteric-exclusion filter never fires on two of the three scoreable apo inputs, and this is why BCR-ABL1's #5 hit is the gatekeeper residue, not a distal one. `func_ligand` excludes active-site contacts by exact ligand-code match against the input structure; BCR-ABL1's declared code is NIL (nilotinib), present only in the holo structure 5M04 and absent from the apo input 1OPL (which carries MYR/P16 instead), and cardiac myosin's list omits V04, the ADP-vanadate transition-state mimic sitting in 8QYP beside the ADP that is listed. The lookup fails closed rather than erroring, so a stale or incomplete code produces no signal that exclusion did not happen. Named here rather than fixed: with under 36 hours to the deadline, re-running invalidates every number already in this package. TASK-0386 tracks the finding; TASK-0391 tracks the fix, scoped for after submission.

target	structure	#1	#2	#3	#4	#5
KRAS_G12C	4LDJ	31	122	33	121	29
BCR_ABL1	10PL	402	311	310	301	338
CARDIAC_MYOSIN	8QYP	682	683	681	680	133
MYC_MAX	1NKP	943	246	925	226	243

target	#1	#2	#3	#4	#5
KRAS_G12C	31	122	33	121	29
BCR_ABL1	321	230	229	220	257
CARDIAC_MYOSIN	611	612	610	609	94
MYC_MAX	46	132	28	112	129

3. Methodological report, per target

KRAS_G12C — 4LDJ

BCR_ABL1 — 10PL

CARDIAC_MYOSIN — 8QYP

MYC_MAX — 1NKP

Our own configuration records that this target has no folded-state allosteric pocket (`allosteric_pocket_exists: false`, `config/targets.yaml` — Myc/Max are IDPs with no surface pocket in the folded dimer); we supply the five because the challenge requires them, and we report that our best candidate site scores 0.161 druggability against our own 0.5 fpocket druggable/non-druggable threshold.

Provenance

Produced by the pipeline at github.com/Dussema-t/Quantum_allosteric-scanner (branch `bartosz`), published 2026-09-15. The five residues in each list match the Concept Proposal's own table exactly — checked, not assumed.

target	AUC	floor (trivial baseline)	paired score - floor, 95% CI	detectable Δ AUC
KRAS_G12C	0.514	0.529 (hop_from_seed)	-0.015 [-0.100, +0.087]	0.093
BCR_ABL1	0.541	0.503 (hop_from_seed)	+0.038 [-0.057, +0.130]	0.094
CARDIAC_MYOSIN	0.548	0.454 (hop_from_seed)	+0.095 [-0.008, +0.201]	0.104

target	→ validated pocket	→ orthosteric ligand	top-site Jaccard
KRAS_G12C (4LDJ)	1.3–14.2 Å	4.3–8.6 Å (GDP·Mg)	0.077 (3/20 overlap)
BCR_ABL1 (10PL)	16.7–26.0 Å	4.2–11.5 Å (P16)	0.000 (0/56)
CARDIAC_MYOSIN (8QYP)	16.6–26.1 Å	6.6–11.8 Å (ADP·VO4·Mg)	0.000 (0/85)

Target: KRAS_G12C | Structure: 4LDJ (apo)

HEADLINE VERDICT

[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

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AUC_apo_Hnew_default      : 0.514
AUC_apo_H10_baseline      : 0.346
AUC_apo_Hnew_optimised    : 0.514
AUC_holo_Hnew_optimised   : N/A
AUC_ctqw_mean             : 0.514
AUC_heat_mean             : 0.342
most_impactful_term        : V_M
least_impactful_term       : V_T
mean_rho_apo_holo         : N/A
mean_jacc20               : N/A
coherence_auc_range       : N/A
coherence_auc_at_gamma0   : N/A
coherence_classification   : N/A

_diagnosis                 : NO_SIGNAL_IN_APO
score 95% CI               : 0.514 [0.348, 0.698] (95% block-bootstrap CI)
floor 95% CI               : 0.529 [0.346, 0.716] (95% block-bootstrap CI)
CIs overlap?               : YES -- statistically indistinguishable from the floor
                           at this confidence level

```

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.168 (meaningful). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.171 (CTQW genuinely helps). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 3) Most/least impactful potential terms: most = V_M, least = V_T.

Target: BCR_ABL1 | Structure: 1OPL (apo)

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HEADLINE VERDICT

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[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

AUC_apo_Hnew_default	:	0.541
AUC_apo_H10_baseline	:	0.520
AUC_apo_Hnew_optimised	:	0.541
AUC_holo_Hnew_optimised	:	N/A
AUC_ctqw_mean	:	0.541
AUC_heat_mean	:	0.664
most_impactful_term	:	V_B
least_impactful_term	:	V_C
mean_rho_apo_holo	:	N/A
mean_jacc20	:	N/A
coherence_auc_range	:	N/A
coherence_auc_at_gamma0	:	N/A
coherence_classification	:	N/A
_diagnosis	:	NO_SIGNAL_IN_APO
score 95% CI	:	0.541 [0.375, 0.689] (95% block-bootstrap CI)
floor 95% CI	:	0.503 [0.350, 0.638] (95% block-bootstrap CI)
CIs overlap?	:	YES -- statistically indistinguishable from the floor
at this confidence level	:	

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.020 (marginal). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = -0.123 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
- 3) Most/least impactful potential terms: most = V_B, least = V_C.

Target: CARDIAC_MYOSIN | Structure: 8QYP (apo, Bos taurus)

HEADLINE VERDICT

[NOTE] AUC values below use the decoherent/time-averaged CTQW limit (a spectral overlap between source and residue eigenvector components), not a coherent quantum-walk snapshot -- all phase information is averaged out by construction. See TASK-0097 / REVIEW-2026-07-13 finding P2-B.

AUC_apo_Hnew_default	:	0.548
AUC_apo_H10_baseline	:	0.468
AUC_apo_Hnew_optimised	:	0.548
AUC_holo_Hnew_optimised	:	N/A
AUC_ctqw_mean	:	0.548
AUC_heat_mean	:	0.514
most_impactful_term	:	V_B
least_impactful_term	:	V_R
mean_rho_apo_holo	:	N/A
mean_jacc20	:	N/A
coherence_auc_range	:	N/A
coherence_auc_at_gamma0	:	N/A
coherence_classification	:	N/A
_diagnosis	:	NO_SIGNAL_IN_APO
score 95% CI	:	0.548 [0.372, 0.774] (95% block-bootstrap CI)
floor 95% CI	:	0.454 [0.301, 0.671] (95% block-bootstrap CI)
CIs overlap?	:	YES -- statistically indistinguishable from the floor
at this confidence level		

DECISION-SUPPORT RECOMMENDATION

- 1) Operator gain over H10 baseline (apo): DAUC = +0.081 (meaningful). -- but not statistically distinguishable from the trivial floor at 95% CI (see _diagnosis above); read this point estimate as a lead, not an established gain
- 2) CTQW vs ground-state relaxation on H_new (same operator, not a classical-diffusion comparison -- see TASK-0095): DAUC = +0.035 (within graph-kernel noise -- CTQW does NOT add biological information beyond the operator).
- 3) Most/least impactful potential terms: most = V_B, least = V_R.

=== MYC_MAX -- NO GROUND TRUTH ===

No AUC, no ceiling, and no proximity-floor check are computed or reported for this target: no holo/bound structure exists for this allosteric question.

This is this target's own documented status (config/targets.yaml), not a scoring failure or an omission -- per this project's own convention, an honest NO is reported explicitly, not silently dropped.

Consensus prediction across 4 independent operators (H_new_default, H10_disorder_suppressed, H2_combinatorial_laplacian, H14_anm_pinv_trace), top-5:

1. residue 943 -- 4/4 operators agree (top-5), mean occupancy=0.0698
2. residue 246 -- 3/4 operators agree (top-5), mean occupancy=0.0599
3. residue 925 -- 3/4 operators agree (top-5), mean occupancy=0.0494
4. residue 226 -- 2/4 operators agree (top-5), mean occupancy=0.0413
5. residue 243 -- 2/4 operators agree (top-5), mean occupancy=0.0405

Confidence: unverified -- at least one residue appears in all 4 operators' own top-5, but operator agreement is not validated confidence: no ground truth exists for this target to check against, and the Concept Proposal's own §2 measures these operators as correlated distance detectors, not independent evidence

Theoretical docking viability (fpocket):

pocket 1: score=0.305, druggability_score=0.161
pocket 2: score=0.112, druggability_score=0.023
pocket 3: score=0.091, druggability_score=0.007
pocket 4: score=0.081, druggability_score=0.158
pocket 5: score=0.054, druggability_score=0.0